

AUTOMOTIVE DIGITAL MARKETPLACE

MVP DEVELOPMENT PROPOSAL

React + Laravel + MySQL

PROPOSED DEVELOPMENT INVESTMENT

₱450,000

Estimated Development Period: 12 Weeks

1. Executive Summary

This proposal covers a lean but commercially useful MVP for an automotive buy-and-sell marketplace. The platform will connect buyers, private sellers, dealers, and parts sellers through a responsive React web application powered by a Laravel backend and MySQL database. The MVP includes five differentiating capabilities: a 360° vehicle viewer, AI-assisted vehicle recommendations, a basic car-parts marketplace, advanced MVP analytics, and real-time WebSocket chat.

The total proposed development investment is ₱450,000. The scope is intentionally optimized for an MVP launch. Enterprise-grade integrations, custom machine-learning development, high-volume infrastructure, and third-party licensing are excluded.

2. Project Objectives

- Launch a functional online marketplace for vehicle buyers and sellers.
- Allow sellers and dealers to publish and manage vehicle listings.
- Allow buyers to search, filter, save, inquire about, and communicate regarding vehicles.
- Provide an integrated basic marketplace for automotive parts.
- Differentiate listings through 360° vehicle presentation.
- Provide personalized vehicle recommendations using an MVP recommendation engine.
- Enable real-time buyer/seller/dealer communication.
- Provide administrators with operational dashboards and analytics.
- Create a scalable foundation for future mobile apps and advanced automotive services.

3. Technology Stack

Layer	Technology
Frontend	React.js
Backend	Laravel / PHP
Database	MySQL 8
Real-Time Chat	Laravel-compatible WebSocket implementation
Authentication	Laravel Sanctum
Storage	AWS S3 or equivalent / server storage
Web Server	Nginx
Version Control	GitHub
Deployment	Linux VPS / Cloud environment
AI	MVP recommendation engine and/or approved AI API

4. MVP Scope

Module	Included Features
Buyer Portal	Registration, profile, browse cars, search, filters, vehicle details, favorites, recommendations, inquiries, offers, appointments, parts browsing/order, chat
Seller Portal	Registration, profile, vehicle listing, photos, 360° media, pricing, publish/unpublish, inquiries, offers, appointments, chat
Dealer Portal	Dealer profile, inventory, basic staff access, leads, appointments, dashboard
Parts Marketplace	Categories, product catalog, product details, search, stock, cart, checkout/request order, seller products, basic order management
360° Vehicle Viewer	360° image upload, interactive rotation, zoom, responsive viewer
AI Recommendations	Recommendations based on viewed/favorited/searched vehicles, budget, category and brand preferences

Real-Time Chat	Buyer-seller/dealer conversations, WebSocket messaging, read/unread, timestamps, basic image sharing
Analytics	Listings, views, favorites, inquiries, appointments, users, popular brands/models, parts orders and sales summaries
Admin Panel	Users, sellers, dealers, vehicles, listings, parts, products, orders, categories, approvals, reports, roles and audit logs

5. Buyer Features

- Secure registration and login
- Buyer profile
- Vehicle marketplace
- Search and basic/advanced filters
- Vehicle details with photos and video
- 360° vehicle viewer
- Favorites and saved vehicles
- AI-assisted vehicle recommendations
- Send inquiry
- Make offer
- Real-time chat
- Book vehicle viewing/test drive
- Browse automotive parts
- Cart and basic parts ordering
- Order/inquiry history
- Notifications

6. Seller & Dealer Features

- Seller registration and profile
- Dealer registration and profile
- Add and edit vehicle listings
- Upload multiple photos
- Upload 360° vehicle images
- Vehicle specifications, mileage, price and location
- Publish/unpublish listings
- Manage inquiries
- Manage offers
- Real-time buyer communication
- Appointment/test-drive management
- Basic listing performance analytics
- Dealer inventory management

7. 360° Vehicle Viewer

The MVP will provide an image-based interactive 360° viewer. Sellers upload a sequence of vehicle images and buyers can rotate and zoom the vehicle.

- 360° image sequence upload
- Interactive drag/swipe rotation
- Zoom

- Mobile-responsive viewer
- Viewer embedded on vehicle detail page

Client/seller is responsible for providing suitable 360° image sequences. Full 3D modeling, photogrammetry, or automated 360° capture is outside the MVP.

8. AI Recommendation Engine

The MVP will use a lightweight recommendation approach rather than training a proprietary machine-learning model. Recommendations may consider buyer searches, viewed vehicles, favorites, budget, brand, model, vehicle category and location.

- Recommended for You
- Similar Vehicles
- Vehicles Within Budget
- Recently Viewed / Related Vehicles

Paid AI/API usage fees and development of a custom machine-learning model are excluded unless separately approved.

9. Car Parts Marketplace

- Parts categories and brands
- Product catalog
- Product photos and descriptions
- Price and stock
- Product search
- Vehicle-compatible product information where supplied
- Shopping cart
- Basic checkout/order request
- Buyer order history
- Parts seller product management
- Parts seller inventory
- Admin order and product management

10. Advanced MVP Analytics

- Total registered buyers, sellers and dealers
- Active vehicle listings
- Vehicle views
- Favorites
- Inquiries and offers
- Appointments
- Popular brands and models
- Listing performance
- Parts products and orders
- Basic sales summaries
- Date-range dashboard filtering

11. Real-Time WebSocket Chat

- Buyer ↔ Seller chat
- Buyer ↔ Dealer chat
- Real-time message delivery
- Conversation history
- Read/unread state
- Timestamps
- Basic image sharing
- Online/availability indicator where supported
- Chat notifications

12. Administrator Panel

- Dashboard
- Buyer management
- Seller management
- Dealer management
- Vehicle management
- Listing approval/moderation
- Brand/model/category management
- Parts/product management
- Order management
- Appointment monitoring
- Basic chat moderation
- Analytics and reports
- Admin users and permissions
- Audit/activity logs
- System settings

13. Project Cost

Cost Component	Budget
UI/UX Design	₱30,000
React Web Application	₱65,000
Laravel Backend/API	₱85,000
MySQL Database	₱25,000
Admin Panel	₱45,000
Buyer & Seller Modules	₱35,000
360° Vehicle Viewer	₱20,000
AI Recommendations	₱20,000
Car Parts Marketplace	₱45,000
Advanced Analytics	₱20,000
Real-Time WebSocket Chat	₱25,000
QA, Deployment & Security	₱20,000
Project Management	₱15,000
TOTAL	₱450,000

14. Suggested Payment Schedule

Milestone	%	Amount
Contract Signing / Kickoff	20%	₱90,000

UI/UX & Database/Architecture Approval	15%	₱67,500
Core React + Laravel + Admin Development	25%	₱112,500
Marketplace Features + Chat + AI + 360°	20%	₱90,000
UAT / Bug Fixing	10%	₱45,000
Production Launch & Handover	10%	₱45,000
TOTAL	100%	₱450,000

15. Estimated Timeline

Period	Activities
Weeks 1–2	Requirements, UI/UX, database, architecture and project setup
Weeks 3–4	Laravel backend, authentication, users, vehicle master data, listings
Weeks 5–6	React marketplace, seller/dealer modules, search/filter, vehicle details
Weeks 7–8	Parts marketplace, 360° viewer, recommendations, analytics
Weeks 9–10	WebSocket chat, notifications, integration and security work
Weeks 11–12	QA, UAT and bug fixing, Final deployment, documentation and handover

16. Explicit MVP Limitations

To maintain the ₱450,000 budget, the following are not included:

- Dedicated React Native Android/iOS applications
- VIN scanner and paid VIN data lookup
- Carfax/AutoCheck-type vehicle history services
- Custom AI/ML model training
- AI damage detection
- Full 3D vehicle modeling or photogrammetry
- Financing/bank integrations
- Insurance integrations
- Escrow services
- Advanced vehicle inspection network
- Large-scale dealer DMS/API synchronization
- Enterprise BI/data warehouse
- High-availability multi-region infrastructure
- Complex multi-country tax/regulatory implementation

17. Third-Party Costs

Third-party costs are separate from the ₱450,000 development fee and may include:

- AWS/cloud hosting
- Domain and SSL
- Payment gateway transaction/service fees
- SMS/OTP
- Email delivery
- Maps/geolocation APIs
- AI/API usage

- Vehicle data/history licensing
- App Store/Google Play fees
- External software licenses

18. Security

- HTTPS
- Secure password hashing
- Laravel authorization and role-based access
- Server-side validation
- File upload validation
- Rate limiting for sensitive endpoints
- Audit logging
- Database backup configuration
- Environment-based secrets
- Development/staging/production separation where practical

19. Future Expansion Roadmap

Future Phase	Potential Features
Phase 2 – Mobile	React Native buyer/seller apps, push notifications, mobile-first workflows
Phase 3 – Automotive Intelligence	VIN lookup, vehicle history, valuation, richer recommendations, AI services
Phase 4 – Parts Expansion	Multi-vendor parts, delivery integrations, supplier APIs, advanced inventory
Phase 5 – Financial & Services	Financing, insurance, inspection, escrow and partner integrations
Phase 6 – Enterprise	Dealer API/DMS synchronization, advanced BI, high-scale infrastructure

20. Maintenance & Support

Post-launch support may be offered under a separate maintenance agreement. A suggested starting range is ₱50,000–₱100,000 per month depending on support hours, SLA, infrastructure responsibilities and ongoing enhancements.

21. Change Request Policy

The ₱450,000 quotation is based on the MVP scope in this proposal. New modules, major workflow changes, additional integrations, custom AI development, or substantial changes after approval will be evaluated separately and may affect cost and delivery schedule.

22. Assumptions

- Client provides branding, logos, content, initial vehicle/parts information and business rules.
- Client provides access to required third-party providers where integrations are approved.
- Initial launch is for one primary market and currency.
- Third-party fees and regulatory/legal expenses are paid by the client.
- 360° media is supplied by sellers/client in a supported image format.
- Final UI/UX designs are approved before corresponding development begins.

- Payment workflows depend on the capabilities and approval of the selected payment provider.

23. Acceptance & Approval

CLIENT / COMPANY: _____

AUTHORIZED REPRESENTATIVE: _____

SIGNATURE: _____

DATE: _____

DEVELOPMENT PROVIDER: _____

AUTHORIZED REPRESENTATIVE: _____

SIGNATURE: _____

DATE: _____